Question Bank – Laplace Transform

Laplace Transform:

Type-I

Find the Laplace transform of $f(t)$, where

1. $f(t) = t, 0 < t < 1/2\,;\ f(t) = t - 1, 1/2 < t < 1;\quad f(t) = 0, t > 1$

2. $f(t) = \sin 2t, 0 < t < \pi;\quad f(t) = 0,\ t > \pi$

3. $f(t) = t, 0 < t < 3;\quad f(t) = 6,\ \ t > 3$

4. $f(t) = \cos t,\ \ 0 < t < 2\pi;\quad f(t) = 0,\ \ t > 2\pi$

5. $f(t) = t^2, 0 < t < 1;\quad f(t) = 1,\ t > 1$

Type -II

Find the Laplace transforms of

1. $2t^3 + cos\,4t + e^{-5t}$ **2.** $e^{2t} + 5t^3 - \sin 2t \cos 3t$ **3.** $\cos(wt + \beta)$

4. $\sqrt{1 + \sin t}$ **5.** $\left(\sqrt{t} - 1\right)^2$ **6.** $\left(\sqrt{t} \pm \frac{1}{\sqrt{t}}\right)^3$ **7.** $3\mathrm{t}^2 + \mathrm{e}^{-\mathrm{t}} + \sin^3 2\mathrm{t}$

8. $sin^4 t$ **9.** $cos^5 t$ **10.** $\cosh^5 t$ **11.** $sin\, t \cdot sin\, 2t \cdot sin\, 3t$

12. If $L\left[sin\sqrt{t}\right] = \frac{\sqrt{\pi}}{2s\sqrt{s}}.e^{-1/4s}$, find $L\left[\sin 2\sqrt{t}\right]$ **13.** If $L[f(t)] = \frac{8(s-3)}{(s^2-6s+25)^2}$, find $L[f(2t)]$

14. If $L[f(t)] = \frac{20-4s}{s^2-4s+20}, find\ L[f(3t)]$

Type –III

Find the Laplace transforms of

1. $e^{-4t} \cos ht \sin t$

2. $e^{-3t} \cos h\ 4t \sin 3t$

3. $e^{2t}(1+t)^2$

4. $e^{2t} sin^4 t$

5. $e^{-2t} \cos h\ t \sin t$

6. If $L[f(t)] = \frac{1}{s(s^2+1)}$, find $L[e^{-t} f(2t)]$

7. $e^{2t} \cos 2t \cos t$

8. $\cos at \cosh at$

9. $t^n e^{-at} + sin^3 t$

Type-IV

Find the Laplace transforms of

1. $te^{-t} \cos h2t$ **2.** $t\, e^{3t} \sin 4t$ **3.** $t\sqrt{1 - \sin t}$ **4.** $t\sqrt{1 + \sin 2\, t}$

5. $t\, e^{3t} erf\sqrt{t}$ **6.** $t\left(\frac{\sin t}{e^t}\right)^2$ **7.** If $L[\mathrm{erf}\sqrt{t}] = \frac{1}{s\sqrt{s+1}}$, find $L[t\ \mathrm{erf}\, 2\sqrt{t}]$

8. If $L[f(t)] = \frac{s+3}{s^2+s+1}, find\ L[t\ f(2t)]$ **9.** $t\ e^t \sin 2t \cos t$ **10.** $te^{-2t} \sin h4t$

11. $t\cos(wt - \alpha)$ **12.** $t^2 \sin 3t$ **13.** $(t + \sin 2t)^2$ **14.** $(t \sin h\ 2\ t)^2$

Type –V

Find the Laplace transforms of

1. $\frac{1}{t}(1 - e^{2t})$ **2.** $\frac{1}{t}(1 - \cos t)$ **3.** $\frac{\sin t}{t}$ **4.** $\frac{e^{2t} \sin t}{t}$ **5.** $\frac{\cos h\ 2t \sin 2t}{t}$ **6.** $\frac{1}{t}[e^{2t} sin^3 t]$

7. $\frac{2sin\, t \sin 2t}{t}$ **8.** $\frac{\sin h\ at}{t}$ **9.** $\frac{sin^2 t}{t}$ **10.** $\frac{1}{t}(\cos at - \cos bt)$

11. $e^{-t} \frac{\sin t \sin 5t}{t}$ **12.** $\frac{(1-\cos 2t)}{e^{2t} \cdot t}$

Type-VI

Find the Laplace transforms of

1. Given $f(t) = \begin{cases} t, & 0 \le t < 3 \\ 6, & t > 3 \end{cases}$, Find $L[f(t)]$ and also $L[f'(t)]$

2. Given $f(t) = \begin{cases} 2t, & 0 \le t \le 1 \\ t, & t > 1 \end{cases}$, Find $L[f(t)]$ and also $L[f'(t)]$, $L[f''(t)]$

3. Given $f(t) = \begin{cases} t^2, & 0 \le t \le 1 \\ 0, & t > 1 \end{cases}$, Find $L[f''(t)]$

4. Given $L(sin\sqrt{t}) = \frac{\sqrt{\pi}}{2s^{3/2}} . e^{-(1/4s)}$ Prove that $L\left(\frac{cos\sqrt{t}}{\sqrt{t}}\right) = \sqrt{\frac{\pi}{s}} . e^{-(1/4s)}$

Type -VII

Find the Laplace transforms of

1. $\int_0^t \frac{1-e^{au}}{u} du$

2. $\int_0^t u^2 \sin u \, du.$

3. $\int_0^t ue^{-3u} \sin 4u \, du$

4. $\int_0^t \frac{e^u \sin u}{u} du$

5. $e^{-3t} \int_0^t u \sin 3u \, du$

6. $\int_0^t e^{-u}. u^4 du$

7. $\int_0^t u. e^{-2u} \sin 3u \, du$

8. $\int_0^t u. e^{-3u} sin^2 \, u \, du$

9. $\int_0^t \int_0^t t \cdot \sin t \, (dt)^2$

10. $\int_0^t \int_0^t \int_0^t \cos 2t \, (dt)^3$

11. $t \int_0^t e^{-4u} \sin 3u \, du$

12. $e^{-t} \int_0^t \frac{\sin u}{u} du$

Type-VIII

Evaluate the following integrals using Laplace transform.

1. $\int_0^\infty e^{-3t}. cos^3 t \, dt$

2. If $\int_0^\infty e^{-2t} \sin(t+\alpha)\cos(t-\alpha) \, dt = \frac{3}{8}$, find α

3. $\int_0^\infty e^{-3t} t \cos t \cdot dt$

4. $\int_0^\infty e^{-2t} t^3 \sin t \, dt$

5. $\int_0^\infty e^{-2t} t \sin^2 t \, dt$

6. $\int_0^\infty e^{-3t} t^2 \sinh 2t \, dt$

7. $\int_0^\infty \frac{\sin at}{t} dt$

8. $\int_0^\infty \frac{e^{-t}.sin\sqrt{3}.t}{t} dt$

9. $\int_0^\infty \frac{e^{-at}-e^{-bt}}{t} dt$

10. $\int_0^\infty \frac{e^{-t}-e^{-3t}}{t} dt$

11. $\int_0^\infty e^{-t} \left(\frac{\cos at - \cos bt}{t}\right) dt$

12. $\int_0^\infty \frac{\cos 6t - \cos 4t}{t} dt$

13. $\int_0^\infty e^{-t}. \frac{(\cos 3t - \cos 2t)}{t} dt$

14. $\int_0^\infty e^{-\sqrt{2}t} \frac{\sin t \sin ht}{t} dt$

15. $\int_0^\infty e^{-2t} \sin ht. \frac{\sin t}{t} dt$

16. $\int_0^\infty e^{-t} \frac{(1-\cos 2t)}{2t} dt$

17. $\int_0^\infty e^{-2t}. \frac{\sin ht}{t} dt$

18. If $L(erf\sqrt{t}) = \frac{1}{s\sqrt{s+1}}$, find $\int_0^\infty e^{-5t} erf(2\sqrt{t}) dt$

19. $\int_0^\infty e^{-t} \int_0^t \frac{\sin u}{u} du \, dt.$

20. $\int_0^\infty e^{-2t} \left(\int_0^t \frac{e^{-u} \sin u}{u} du\right) dt$

21. $\int_0^\infty e^{-2t} \left[\int_0^t \left(\frac{1-e^{-u}}{u}\right) du\right] dt$

22. $\int_0^\infty e^{-4t} \left[\cosh t \int_0^t e^u \cosh u \, du\right] dt$

23. $\int_0^\infty e^{-t} \left(t \int_0^t e^{-4u} \cos u \, du\right) dt$

24. $\int_0^\infty e^{-t} \left(\frac{1}{t} \int_0^t e^{-u} \sin u \, du\right) dt$

Inverse Laplace Transform :

Type-I

Find the inverse Laplace transform of

1. $\frac{1}{(s+2)^4}$ **2.** $\frac{6s-4}{s^2-4s+20}$ **3.** $\frac{s}{s^2+2s+2}$

4. $\frac{s}{(2s+1)^2}$ **5.** $\frac{s+4}{(s^2-1)(s+1)}$ **6.** $\frac{2s^2-6s+5}{s^3-6s^2+11s-6}$

7. $\frac{1}{(s-2)^2(s+3)}$ **8.** $\frac{1}{s^3(s-1)}$ **9.** $\frac{1}{s^2(s+2)}$

10. $\frac{5s+3}{(s-1)(s^2+2s+5)}$ **11.** $\frac{1}{s^3+1}$ **12.** $\frac{s+29}{(s+4)(s^2+9)}$

13. $\frac{2s-1}{(s^4+s^2+1)}$ **14.** $\frac{a(s^2+2a^2)}{s^4+4a^4}$ **15.** $\frac{s}{(s+1)^2(s^2+1)}$

16. $\frac{s+2}{(s+3)(s+1)^3}$ **17.** $\frac{s}{(s^2+1)(s^2+4)(s^2+9)}$ **18.** $\frac{s}{(s^2+1)(s^2+4)}$

19. $\frac{s^2}{(s+4)^3}$ **20.** $\frac{2s^2-1}{(s^2+1)(s^2+4)}$

Type-II

Find the inverse Laplace transforms of the following functions by using convolution theorem

1. $\frac{1}{(s-2)(s+2)^2}$ **2.** $\frac{s^2}{(s^2+2^2)^2}$ **3.** $\frac{s}{(s^2+1)^2}$

4. $\frac{1}{s^2(s+1)^2}$ **5.** $\frac{s^2}{(s^2+1)(s^2+4)}$ **6.** $\frac{s}{(s^2+4)(s^2+1)}$

7. $\frac{1}{s^2-s-6}$ **8.** $\frac{16}{(s-2)(s+2)}$ **9.** $\frac{(s+5)^2}{(s^2+10s+16)^2}$

10. $\frac{s+3}{(s^2+6s+13)^2}$ **11.** $\frac{s}{s^4+8s^2+16}$ **12.** $\frac{(s-1)^2}{(s^2-2s+5)^2}$

13. $\frac{s+1}{(s^2+2s+2)^2}$

Type-III

Find the inverse Laplace transform of

1. $log\left[\frac{s^2-4}{(s-3)^2}\right]$ **2.** $log\left[1+\frac{1}{s^2}\right]$ **3.** $\frac{1}{2}log\left(1-\frac{a^2}{s^2}\right)$

4. $log\sqrt{\frac{s^2+1}{s^2}}$ **5.** $log\left[\frac{s^2+a^2}{(s+b)^2}\right]$ **6.** $log\left(1+\frac{a}{s}\right)$

7. $log\sqrt{\frac{s^2+a^2}{s^2}}$ **8.** $log\left(1-\frac{1}{s^2}\right)$ **9.** $tan^{-1}(s+1)$

10. $tan^{-1}\frac{1}{s}$ **11.** $cot^{-1}s$ **12.** $tan^{-1}\left(\frac{s}{2}\right)$

13. $\frac{s}{(s^2+a^2)^2}$ **14.** $\frac{s+3}{(s^2+6s+13)^2}$

Using convolution theorem, prove that **(15 to 19)**

15. $L^{-1}\left[\frac{1}{s}log\left(\frac{s+3}{s+4}\right)\right] = \int_0^t \frac{e^{-4u}-e^{-3u}}{u}du$ **16.** $L^{-1}\left[\frac{1}{s}.log\left(\frac{s+2}{s+3}\right)\right] = \int_0^t\left(\frac{e^{-3u}-e^{-2u}}{u}\right)du$

17. $L^{-1}\left[\frac{1}{s}tan^{-1}\left(\frac{s+a}{b}\right)\right] = \int_0^t -\frac{e^{-au}}{u}\sin bu\ du$ **18.** $L^{-1}\left[\frac{1}{s}\tan^{-1}\frac{2}{s}\right] = \int_0^t \frac{1}{u}\cdot\sin 2u\,du$

19. $L^{-1}\left[\frac{1}{s}\log\left(1+\frac{4}{s^2}\right)\right] = \int_0^t \frac{2}{u}[1-\cos 2u]\,du$

Find the inverse Laplace transform of

20. $\frac{s^2}{(s^2+1)^2}$

21. $\frac{(s+2)^2}{(s^2+4s+8)^2}$

22. $\frac{s^2}{(s+1)^3}$

23. $\frac{s^2}{(s^2-1)^2}$

24. $\frac{1}{s^3(s^2+1)}$

25. $\frac{54}{s^3(s-3)}$

26. $\frac{1}{s(s^2+9)}$

27. $\frac{1}{s^2(s+2)}$

Type –IV

Find the Laplace transform of

1. $f(t) = \sin 2t, 0 < t < \frac{\pi}{2}$, $f(t) = 0 , \frac{\pi}{2} < t < \pi$ *and* $f(t) = f(t+\pi)$
2. $f(t) = |\sin p\, t|, t \geq 0$
3. $f(t) = \frac{t}{a}, 0 < t \leq a;\ f(t) = \frac{1}{a}(2a - t), a < t < 2a$ and $f(t) = f(t+2a)$
4. $f(t) = \begin{cases} E, & 0 \leq t \leq (p/2) \\ -E, & (p/2) \leq t \leq p, \end{cases} \qquad f(t+p) = f(t)$

Type –V

Find the Laplace transform of

1. $\sin t\, H\,(t - \pi)$
2. $t^2 H(t-2)$
3. $e^{-t} sint\, H(t-\pi)$
4. $(1 + 2t - t^2 + t^3)H(t-1)$
5. $e^{-t} \cos t\, H(t-\pi)$
6. $(1 + 3t - 4t^2 + 2t^3)H(t-3)$

Express the following functions as Heaviside's unit step functions and find their Laplace transforms.

7. $f(t) = \begin{cases} (t-3)^4, & t > 3 \\ 0, & 0 < t < 3 \end{cases}$
8. $f(t) = \begin{cases} 0, & 0 < t < 1 \\ t^2, & 1 < t < 3 \\ 0, & t > 3 \end{cases}$
9. $f(t) = \begin{cases} e^t \sin t\,, & 0 < t < \pi \\ e^t \cos t\,, & t > \pi \end{cases}$
10. $f(t) = \begin{cases} \cos t\,, & 0 < t < \pi/2 \\ \sin t\,, & t > \pi/2 \end{cases}$
11. $f(t) = \begin{cases} \sin t\,, & 0 < t < \pi \\ \sin 2t\,, & \pi < t < 2\pi \\ \sin 3t\,, & t > 2\pi \end{cases}$

Find the inverse Laplace transform of

12. $e^{-s}\frac{(1+\sqrt{s})}{s^3}$
13. $\frac{e^{-4s}}{\sqrt{2s+7}}$
14. $\frac{s.e^{-2s}}{s^2+2s+2}$
15. $\frac{(s+1)e^{-2s}}{s^2+2s+2}$
16. $\frac{se^{-s/2}+\pi e^{-s}}{s^2+\pi^2}$
17. $\frac{e^{-\pi s}}{s^2(s^2+1)}$
18. $e^{-2s} \sin 2$
19. $e^{-3s} \cos 3$

Using Laplace transform evaluate the following integrals

20. $\int_0^{\infty} e^{-2t}(1 + t + t^2)H(t-3)\, dt$
21. $\int_0^{\infty} e^{-t}(1 + 3t + t^2)H(t-2)\, dt$

Type –VI

Using Laplace transform solve the following differential equations with the given conditions.

1. $3\frac{dy}{dt} + 2y = e^{3t}, y = 1\ at\ t = 0$
2. $\frac{dx}{dt} + y = \sin t,\ \frac{dy}{dt} + x = \cos t$ where $x = 0, y = 2\ at\ t = 0$

3. $2y'' + 5y' + 2y = e^{-2t}, y(0) = y'(0) = 1$

4. $\frac{d^2y}{dt^2} + \frac{dy}{dt} - 2y = e^{3t}, given\ y(0) = 0\ and\ y'(0) = 1$

5. $(D^2 - 2D + 1)x = e^t$ with the conditions $x = 2, Dx = -1\ at\ t = 0$

6. $(D + 1)^2 y = 6te^{-t} with\ y(0) = 2, y'(0) = 5$

7. $(D^2 - 2D - 8)y = 4, y(0) = 0\ and\ y'(0) = 1$

8. $\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 8y = 1$ where $y(0) = 0, y'(0) = 1$

9. $\frac{d^2y}{dt^2} + 9y = \cos 2t,\ y(0) = 1\ and\ y(\pi/2) = -1$

10. $(D^2 - D - 2)x = 2(1 + t - t^2); x = 0, Dx = 3\ for\ t = 0$

11. $\frac{d^2y}{dt^2} + 4y = f(t)$, with conditions $y(0) = 0, y'(0) = 1$ and $f(t) = H(t - 2)$

12. $y + \int_0^t y\, dt = 1 - e^{-t}$